PRISM (NeurIPS 2026 Submission 8674) — Rebuttal 分析與草稿

Scores: pCi8 = 3 (conf 4) | G3T9 = 3 (conf 4) | eQL6 = 3 (conf 3) | 8VrD = 4 (conf 3)
AC Meta-Review (LN7U, 2026-07-17, 修訂 07-24): preliminary **Borderline Reject** — 但明確寫出「a strong and convincing rebuttal with additional experiments and careful narrowing of the claims could move the paper closer to borderline accept」，並逐條列出五項可翻盤條件（見 §0.5）。

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0. 局勢判斷與整體策略

Viability: BORDERLINE / PROMISING。值得全力 rebuttal，理由：

1. 沒有任何 reviewer 指出 correctness 的致命錯誤。唯一接近 correctness 的兩點（8VrD 的 empirical vs. population risk、eQL6 的 scale-domination 質疑）都可以用「定理重述 + 澄清」完全解決，不需要改方法。
2. 三位給 3 分的 reviewer，Quality/Originality 都給 3（good），扣分集中在 **Significance=2**（pCi8, G3T9）與 **Clarity=2**（eQL6）——這兩類是 rebuttal 最容易翻轉的類型（補實驗 + 澄清），不是「方法錯了」。
3. 8VrD（4 分）是潛在 advocate，其 review 最深入、要求最具體，把他餵飽可以穩住甚至升 5，並在 AC discussion 幫你說話。
4. 四位的 weakness 高度重疊（bound 太鬆、baseline 不足、teacher-forcing、reference set 敏感度），一組實驗可以同時回應多位，投報率高。
5. **AC** 已把翻盤條件白紙黑字寫出來（§0.5 的 A–E 五項），並明言要「concrete evidence rather than only clarification」。這把 rebuttal 從開放式作文變成有明確評分標準的考卷——照著 A–E 逐項交付證據即可。

優先順序（就分數期望值而言）：**AC** 的五項 **asks** 優先於任何單一 **reviewer**（meta-review 直接決定 accept/reject），其次 8VrD（穩住/加深 advocate）≥ eQL6（技術誤解可澄清，Clarity 2 最好翻）> G3T9 > pCi8。所幸 AC 五項與四位 reviewer 的關切幾乎完全重疊，同一組實驗雙線得分。

建議格式：一則簡短 Global Response（列出新增實驗 E1–E6 與誠實性修訂），加上每位 reviewer 各自獨立、自包含的回覆。字數限制請確認 NeurIPS 2026 規則（通常每則 comment ~5000–6000 字元）[VENUE RULE NEEDS VERIFICATION]。

誠實性原則：以下草稿中所有 `[XX]` / `[RESULT NEEDED]` 都必須以真實跑出的數字填入；跑不出來就改用 narrow concession 的備用句（各點已附）。絕不虛構數字。

0.5 AC Meta-Review（Area Chair LN7U）分析與逐點對應

現狀判讀

AC 在 rebuttal 前就發出 preliminary meta-review（07-17 發、07-24 修訂），這是邀請函而非判決書：AC 花篇幅寫出「What could change the recommendation?」五項具體條件，代表 AC 認為本文可救、且願意被說服。三個關鍵訊號：

1. 「**concrete evidence rather than only clarification**」— AC 預先排除了純澄清式 rebuttal 的效力。P0 實驗（E1–E6）不是加分項，是入場券。
2. 「**careful narrowing of the claims**」被 **AC** 明列為加分行為 — 誠實劃界（empirical-risk 定理重述、teacher-forced 限定、負面 cell 進主文）不會被當成示弱，反而是 AC 點名要的。
3. AC 對弱點的歸納與 §1 的 S1–S6 幾乎一一對應，且 AC 未引入任何新的技術性反駁——沒有新戰場，只需把既定計畫執行到位。

AC 弱點清單 → 本文件章節對應：loose bound（S1）、diagnostic utility（pCi8-W1/W3、G3T9-W1）、baselines（S2）、narrow setting（S3/S4）、regularization 證據不足（S5/S6）、clarity/理論框架（eQL6-W1/W2、8VrD-W1）。

AC 五項翻盤條件（A–E）→ 交付物對應表

AC ask	AC 原文要點	對應實驗/修訂	對應 reviewer	交付形式
A. Clarify usefulness of the bound	明說是 empirical/calibration-set bound 還是 population 陳述；量化 looseness；ranking 之外有無 operational threshold	E5（定理重述 + concentration corollary）+ E6（slack 分佈 + isotonic calibration）	8VrD-W1/W2, pCi8-W2, G3T9-W4	重述後的 Theorem 陳述貼進 global response + slack/calibration 數字表
B. Benchmark-independent use	用小型、與 benchmark 無關的 reference set 仍具預測力；含 size/domain 敏感度	E3（size {8–128} × domain C4/WikiText ablation）	G3T9-W2, 8VrD-Q3	一張 ablation 表：rank correlation vs. reference set 設定

AC ask	AC 原文要點	對應實驗/修訂	對應 reviewer	交付形式
C. Stronger baselines	診斷:CKA、SVCCA、簡單 feature-preserving scores 直接比較;正則化:EWC、SLoRA、CLAIM、ArMA 或 **layer-freezing/replay variants**	E1(相似度 baseline ranking 表)+ E2(EWC、L2-SP、feature-KD,加 **layer-freezing**——AC 明列為可接受替代,成本低)	G3T9-W3, pCi8-W5, 8VrD-W3/Q4	兩張 baseline 表(診斷、正則化),含 seeds 與 target-task 欄
D. Free-running generation	提供 free-run 實驗,或明確把 claim 限定在 teacher-forced	E7(子集 free-run);跑不完就走 AC 給的第二條路:corollary 重述為 teacher-forced only + Limitations 明文	G3T9-W2, 8VrD-W4/Q2	實驗數字,或明確的 claim-narrowing 修訂文字
E. Analyze failures and mixed results	討論 GSM8K、Qwen-BBQ 弱相關;regularizer 無效/惡化的 cases	E9(GSM8K answer-span 分析)+ E10(Table 22 axis-specificity 重詮釋)+ Qwen-BBQ 失敗分析進主文	pCi8-W4, 8VrD-W3/W4	失敗模式分析段落 + 全 cell 聚合統計(不再只報 Llama 平均)

執行判準:A、B、C、E 四項必須有數字;D 允許二選一。五項全交付 →「closer to borderline accept」是 AC 自己寫的承諾。

對 rebuttal 結構的三個直接影響

1. **Global response** 改為以 **AC** 五項為骨架(原建議是以六項新實驗為骨架):開頭一句話點名「we address each of the five points raised in the meta-review with new experiments」,然後 A–E 各一段(每段 2–4 句 + 關鍵數字),結尾一句 claim-narrowing 總結。AC 是 global response 的第一讀者,用他的分類法讓他零成本核對。
2. **E7**(**free-run**)從 **P1** 升級為 **P0-or-narrow**:AC 給了明確的二選一,但「有子集數字」與「只劃界」的說服力差一級;至少跑一個 (family, benchmark) cell。

3. **E2 加入 layer-freezing baseline**（AC 明列，實作成本最低的「stronger baseline」）；SLoRA/CLAIM/ArMA 若 setting 不合，用「已加 EWC + L2-SP + feature-KD + layer-freezing + replay，並在 Related Work 討論四篇」回應即可站得住。

Global Response 草稿（對 AC 五項逐條，貼於 top-level comment）

Global Response — new experiments and revisions addressing the meta-review

We thank the reviewers and the AC. The meta-review distills the reviews into five concrete asks; we address each with new evidence, and we have narrowed our claims where the evidence warrants it.

(A) Usefulness of the bound. Theorem [X] is now restated as a bound on the **empirical risk gap over the reference sample**, with a new concentration corollary (McDiarmid, under the bounded-norm assumption PRISM already measures) bridging to population risk at an additive $O(\sqrt{\log(1/\delta)/N})$ cost. We further quantify looseness directly — slack $B/|\Delta R|$ distribution: median [XX], IQR [XX–XX] — and add a held-out per-family isotonic calibration mapping B to predicted $|\Delta R|$ (MAE [XX] nats), which yields an operational threshold: "$B < \tau \Rightarrow |\Delta R| < \varepsilon$" with precision [XX] on held-out variants. Uncalibrated B is now claimed for ranking only. ([RESULT NEEDED] ×3)

(B) Benchmark-independent use. Recomputing PRISM with a generic [C4/WikiText] reference set (no benchmark overlap) preserves rankings: mean Spearman [XX] vs. [XX] benchmark-specific; sensitivity over {8, 16, 32, 64, 128} sequences and two domains is [XX] (new Appendix [X]). ([RESULT NEEDED])

(C) Stronger baselines. Diagnostics: PRISM vs. linear-CKA / SVCCA / Procrustes distance under the identical ranking protocol — mean r_s [XX] / [XX] / [XX] / [XX] (new Table [X]); beyond correlation, none of these provides risk linkage or axis attribution. Regularization: EWC, L2-SP, feature distillation, and layer-freezing added under the identical protocol, [N] seeds, with target-task performance reported (new Table [X]). ([RESULT NEEDED] ×2)

(D) Free-running generation. On [subset], PRISM computed on model-generated continuations gives [XX] ([RESULT NEEDED]); [fallback: we have restated Corollary [X] as teacher-forced-only and list trajectory-distribution shift as an explicit limitation and open extension].

(E) Failures and mixed results. The revised main text reports **all** correlation cells including Qwen-BBQ (−0.34, −0.66) with all-cell aggregates (mean [XX], median [XX]) and a failure analysis [hypothesis + evidence]; GSM8K is promoted to a first-class limitation with an answer-span variant analysis ([XX] → [XX]); and Table 22's negative regularizer cells are foregrounded — [if E10 holds: PRISM's own axis diagnosis attributes forgetting in those cells to non-shape axes, so the shape regularizer is now presented as axis-targeted, applied when the shape term dominates]. ([RESULT NEEDED] / [VERIFY])

Per-reviewer responses contain the full tables and derivations. We believe these results, together with the narrowed framing — PRISM as a **label-free ranking-and-attribution instrument with an explicitly empirical-risk guarantee** — address the meta-review's assessment, and we respectfully ask the AC and reviewers to reconsider in light of the new evidence.

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1. 跨 Reviewer 共同關切矩陣

#	共同關切	**Reviewers + AC**	**Severity**	回應模式
S1	Bound 絕對值太鬆（loose by orders of magnitude），只能 ranking	pCi8-W2, G3T9-W4, 8VrD-W2/Q1, **AC-A**	Major, ALL+AC	承認 + 量化 slack + 新增 isotonic calibration 實驗 + 收斂 claim
S2	Baseline 不足：CKA/SVCCA 未直接比較；forgetting 只比 replay，缺 EWC / L2-SP / feature-KD / layer-freezing / 近期 CL 方法	pCi8-W5, G3T9-W3, 8VrD-W3/Q4, **AC-C**	Major, 3/4+AC	P0 新實驗（E1, E2）
S3	Teacher-forcing 限制，free-running generation 未驗證	G3T9-W2, 8VrD-W4/Q2, **AC-D**	Major+AC	E7 實驗 or 明確劃界（AC 接受二選一）

#	共同關切	**Reviewers + AC**	**Severity**	回應模式
S4	Reference set 敏感度（大小、domain、是否 benchmark-independent）	G3T9-W2, 8VrD-W3/Q3, **AC-B**	Major+AC	P0 便宜 ablation（E3）
S5	Regularizer 對原任務（target task ）performance 的影響未報告	eQL6-W4/Q3, 8VrD-W3, **AC**（ stability-plasticity tradeoff 未建立）	Major+AC	P0 補表（E2 一併報告）
S6	負面結果需誠實呈現（GSM8K、Qwen-BBQ 負相關、Table 22 forgetting 增加）	pCi8-W4, 8VrD-W3/W4, **AC-E**	Major+AC	主文改寫 + axis-specificity 重新詮釋（E10）+ 失敗分析

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2. Reviewer pCi8（3 分，conf 4；Significance=2）逐點回應

pCi8-W1｜三個診斷結論早已知，貢獻只是統一框架

- 本質：NOVELTY / SIGNIFICANCE。這是拉低 Significance 的主因。Reviewer 已在 Strengths 承認 simplex polarization、exact decomposition 是「real contribution」，所以他不是否定技術，而是否定「發現」。
- 策略：`structural_distinction` + 測量儀器框架。不要否認「現象已知」——同意它，然後指出貢獻是把三個各自孤立的、只有定性描述的現象放進同一個以 **nats** 為單位、可交換比較（**commensurable**）的 **risk bound**，因此第一次能做（i）無標籤的量化歸因（哪個軸主導、佔多少比例）、（ii）跨 variant 的排序、（iii）可微分的介入。已知現象 ≠ 已有測量工具：溫度計沒有發現發燒，但沒有溫度計就無法分診。
- 草稿：

W1 (unified framework vs. new discovery). We agree that each failure mode had been observed *qualitatively* in isolation, and we will cite this prior evidence more prominently. Our claimed contribution is not the discovery of the phenomena but the first **quantitative, label-free instrument that makes them commensurable**: all three axes are expressed in the same units (contribution to a certified CE-risk-gap bound), so for a *given* variant one can compute *which* axis dominates and by how much — something qualitative prior knowledge cannot do. This is what enables (i) ranking without benchmarks, (ii) attribution (Sec. 5.3), and (iii) the differentiable intervention (Sec. 5.4), which the reviewer acknowledges as strengths. The analogy we would offer: the phenomena were known the way "fever" was known before thermometers; PRISM supplies the thermometer and the triage rule. We will sharpen the contribution statement in the Introduction accordingly.

pCi8-W2｜Bound 鬆了幾個數量級，只 calibrated for ranking，無法回答「Q4 夠不夠好」

- 本質：S1（共同關切）。注意陷阱：slack 不是常數倍（B=266.09 vs |ΔR|=0.3658 約 727×；B=23.24 vs 0.0002 約 10^5×），不要宣稱 multiplicative slack 穩定，會被抓。
- 策略：三段式。(a) 承認：distribution-free 的 CE upper bound 在絕對值上必然保守，這是所有此類 bound 的共性（cite 相似性質的 generalization bound 文獻）。(b) 加值：新增 **held-out isotonic / per-family affine calibration**，把 B 映射到預測的 ΔR 並附 error bar，回答「夠不夠好」的操作性問題。(c) 收斂 claim：主文明確標示「calibrated for ranking」的適用範圍。
- 草稿：

  W2 (bound looseness / "is Q4 good enough?"). We agree the raw bound is loose in absolute scale — this is inherent to distribution-free upper bounds and we will state it more prominently (not only in Sec. [X]). To make the bound *operational* beyond ordering, we added a calibration study: on a held-out set of variants we fit a per-family isotonic map from B to the observed |ΔR|; on the remaining variants this calibrated predictor achieves mean absolute error [XX] nats ([RESULT NEEDED]), i.e., after a one-time calibration with [K] benchmarked variants, PRISM answers "is Q4 within ε of the base?" with quantified uncertainty, while still requiring **no labels for every new variant**. We will add this as Sec. [X] / Table [X] and explicitly narrow the uncalibrated claim to ranking.

- 備用句（實驗來不及）：承認 + "we view absolute-scale certification as an open problem and will state this as an explicit limitation; the intended use case is cheap screening of many variants followed by benchmarking only the top-k."

pCi8-W3｜Table 3 顯示 Ω 單獨用 vs. 完整 PRISM 差距很小 → 理論機器太重、增益太小

- 本質：EMPIRICAL_SUPPORT / SIGNIFICANCE。要害是「機器的價值不只在 ranking 增益」。
- 策略：(a) 指出 Ω-only 在 head-divergence 主導的情境（GGUF k-quant 量化 lm_head）原理上不可見——head 項不在 Ω 裡；引用論文中 head 軸主導的 case [VERIFY: 指出具體 table/case，並檢查該情境下 Ω-only 是否確實 misrank；若是，這是最有力的一擊]。(b) 完

整 bound 是推導出 Ω 是正確相似度的理由（回答「為何不是 CKA」），沒有它 Ω 只是又一個 heuristic。(c) 三軸歸因與 regularizer 都來自完整分解，非 Ω-only 可得。
- 草稿：

  W3 (Ω alone vs. full PRISM). Three points. (1) *Ranking:* on backbone-dominant drift (most PTQ cells) Ω alone is indeed competitive — expected, since the shape term dominates there. However, Ω is **blind by construction to the head axis**: in the GGUF k-quant setting where lm_head is quantized (Sec. [X]), the head term is what detects degradation; Ω-only [misranks these variants / is uninformative there: [VERIFY & report the specific cells]]. (2) *Attribution:* the practical output of PRISM is not only a score but *which axis to fix*; Ω-only offers no decomposition. (3) *Principle:* the bound is what derives Ω as the risk-relevant similarity (rather than CKA/SVCCA) and what yields the differentiable regularizer — the "machinery" is the justification, not overhead. We will add an explicit "when is Ω alone enough?" paragraph to Sec. [X] making this scope clear.

pCi8-W4｜GSM8K 相關性明顯較低，長 CoT 稀釋 per-token loss，是結構性限制

- 本質：GENERALIZATION，且指向 reasoning 時代的適用性——這點必須嚴肅承認，不能硬拗。
- 策略：承認結構性 + 精確定位根因（是 **CE-as-target** 與 **final-answer accuracy** 的脫鉤，任何以 CE 為對象的 proxy——包含 perplexity screening——都繼承此限制，並非 PRISM 幾何分解特有）+ 提出緩解（把 reference/評估 token 限縮到 answer span 或以 answer-token 加權重算）+ 若能跑出改善數字最好。
- 草稿：

  W4 (GSM8K / long CoT). We agree this is structural, not an edge case, and we will promote it from Sec. [X] to a first-class limitation. Two clarifications. (1) The root cause is the decoupling between *per-token CE* and *final-answer accuracy* on long CoT — a property of the CE target itself, inherited by any CE-based label-free proxy (including perplexity screening), not specific to our geometric decomposition. (2) It is partially mitigable within PRISM: restricting the reference features to answer-span tokens (re-weighting the feature matrices) raises GSM8K rank correlation from [XX] to [XX] ([RESULT NEEDED]; if not completed: "we outline this answer-span variant and leave its evaluation to the revision"). We will add this analysis and discuss implications for reasoning-heavy models in the Limitations.

pCi8-W5｜引用了 EWC 卻沒比；只比小規模 experience replay 是「方便的選擇」

- 本質：MISSING_BASELINE（S2）。「convenient choice」措辭帶刺，回應要冷靜、直接給數字。

- 策略：P0 實驗 E2：EWC（Fisher diagonal on LoRA params）+ L2-SP + feature-space distillation，同 protocol、多 seed、同時報 target-task 與 forgetting（一次滿足 pCi8-W5、G3T9-W3、8VrD-W3/Q4、eQL6-W4）。
- 草稿：

  W5 (EWC baseline). We completed the requested comparison. Under the identical LoRA-forgetting protocol (same data, rank, steps; [N] seeds), on [benchmark cells]: shape-regularizer [XX], EWC [XX], L2-SP [XX], feature distillation [XX], replay [XX] ([RESULT NEEDED] — report retention and target-task metric per method). [Summarize honestly: where shape-reg wins, ties, loses.] We will add this as Table [X]. We chose replay initially as the strongest *data-access* baseline, but agree EWC and parameter/feature-space regularizers are the canonical comparison and thank the reviewer for pushing on this.

pCi8-W6｜Orthogonal alignment 依賴 near-isometry；激進量化把幾何打壞後假設還成立嗎？

- 本質：CORRECTNESS-adjacent，但其實是對 **bound** 邏輯的誤讀，是很好的澄清機會。
- 策略：關鍵區分——**bound** 的有效性（**validity**）不需要 **near-isometry**；**near-isometry** 只影響緊度（**tightness**）。Bound 對任意 orthogonal alignment 成立（取 inf/任一 O 都是 upper bound）[VERIFY 與論文推導一致：確認 bound 是對所有 O∈O(d) 成立、取 min 只是收緊]。幾何被打壞時，bound 變鬆的方向是保守的（高估 risk gap），對「排序壞 variant 在後」無害且實證上排序仍對（2-bit cell 的 ranking 結果本身就是證據）。再補一個實測：isometry violation（例如 unconstrained Procrustes 最優解偏離 orthogonal 的程度）vs. bit-width 的曲線（E8）。
- 草稿：

  W6 (near-isometry under aggressive quantization). This is an important distinction we will make explicit: **the bound's validity does not require near-isometry — only its tightness does.** Theorem [X] holds for *any* orthogonal alignment (we take the minimizer only to tighten it); when aggressive quantization destroys geometry, the bound loosens in the conservative direction (over-estimating the gap for the most-damaged variants), which preserves ranking — consistent with the observed 2-bit results (Table [X]). Empirically, we now also measure the isometry violation directly: the deviation of the optimal *unconstrained* linear alignment from the orthogonal group grows from [XX] at 8-bit to [XX] at 2-bit ([RESULT NEEDED]), yet rank correlation in those cells remains [XX]. We will add this measurement and the validity-vs-tightness clarification to Sec. [X].

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3. Reviewer G3T9（3 分，conf 4；Significance=2）逐點回應

G3T9-W1｜動機不足：何不直接跑 benchmark？三軸太粗、非 actionable root cause；要求成本分析

- 本質：SIGNIFICANCE 的核心質疑。必須給出（a）具體成本數字、（b）「benchmark 告訴你掉分、PRISM 告訴你修哪裡」的 remediation 對照。
- 策略：做一張 cost table：K 個 variant × 全 benchmark suite（含 GSM8K 等需要 autoregressive decoding 的）GPU-hours vs. PRISM（每 model 一次 forward、32 sequences、無標籤、無 decoding）。再列 axis→action 對照：shape→Hessian-aware reconstruction / 重新量化敏感層；scale→norm/scale re-calibration；head→lm_head 保留 fp16（GGUF 案例：診斷出 head 軸 → 不量化 head → 修復，這就是 actionable root cause 的實例）。強調無標籤 ⇒ 在沒有 benchmark 的私有 domain 也可用。
- 草稿：

  W1 (why not just benchmark? / actionability / cost). We will add the requested cost analysis (new Table [X]). Concretely: ranking [K] PTQ variants over our five benchmarks requires [XX] GPU-hours (dominated by autoregressive decoding for GSM8K/CoT), versus [XX] GPU-minutes for PRISM — one teacher-forced forward pass per model over 32 reference sequences, no labels, no decoding ([RESULT NEEDED]; exact protocol in Appendix [X]). Beyond cost, the two tools answer different questions: benchmarks report *that* and *how much* accuracy dropped; PRISM reports *which component* drifted, which maps to a concrete action — shape → Hessian-aware layer re-quantization, scale → normalization re-calibration, head → keep lm_head in fp16. The GGUF case (Sec. [X]) is an end-to-end instance: PRISM attributed degradation to the head axis; keeping the head unquantized removed it. Finally, PRISM is label-free, so it applies where no benchmark exists (proprietary domains). We will make this positioning explicit in the Introduction.

G3T9-W2｜Teacher-forcing；bound 用 benchmark 本身的輸入算；需要 benchmark-independent reference set 與樣本數 ablation

- 本質：S3 + S4。三個具體要求都合理且便宜，全做。
- 草稿：

  W2 (reference set independence, sample ablation, free-running). Three parts. (1) *Benchmark-independent reference set:* recomputing PRISM with a generic [C4/WikiText] reference set (no overlap with evaluation benchmarks) yields mean Spearman [XX] vs. [XX] for benchmark-specific references ([RESULT NEEDED]) — [state honestly whether rankings are preserved]. (2) *Sample efficiency:* varying the reference set over {8, 16, 32, 64, 128} sequences gives rank correlations [XX / XX /

XX / XX / XX] ([RESULT NEEDED]), stabilizing at [N]. Both will be added as Appendix [X]. (3) *Free-running:* we agree teacher-forcing is a real scope restriction. On a subset we recompute features on model-generated continuations: [XX] ([RESULT NEEDED]; fallback: we will narrow the autoregressive corollary's claim to teacher-forced prefixes and add trajectory-shift as an explicit limitation and future-work term).

G3T9-W3｜要與 CKA / SVCCA 同場直接比較；regularizer 要比 simpler feature-preserving regularizers

- 本質：S2。這是決定性的表——論文定位就是「比 CKA/SVCCA 更好」，缺這張表等於核心 claim 沒有直接證據。P0 最高優先。
- 草稿：

W3 (direct comparison to CKA/SVCCA; simpler regularizers). Agreed — this comparison should be in the paper, and both are now included. (1) *Ranking:* under the identical protocol (same variants, same reference activations), mean Spearman across PTQ cells: PRISM [XX], linear-CKA [XX], SVCCA [XX], Procrustes distance [XX] ([RESULT NEEDED]); per-cell results in new Table [X]. Beyond aggregate correlation, CKA/SVCCA provide no risk link and no axis attribution, so even where correlations are close, they cannot produce Sec. 5.3's diagnoses. (2) *Regularizers:* see the new baseline table (shared with R-pCi8 W5): L2-SP and feature-distillation columns added, with target-task performance reported ([RESULT NEEDED]).

G3T9-W4｜Bound 絕對尺度太鬆 → 只是 heuristic ranking score

- 同 S1，直接引用給 pCi8-W2 的回覆內容（isotonic calibration + claim narrowing）。每個 thread 要自包含，所以把 pCi8-W2 草稿濃縮一段貼入，不要寫 "see Reviewer pCi8"。

G3T9-Q1｜能否用於 full fine-tuning / RLHF？

- 策略：誠實劃界 + 指出一般形式本來就涵蓋 head 變動。Full FT/RLHF：分解定義上仍成立（同架構、同維度即可），但 drift 越遠、orthogonal alignment 越不緊，預期 bound 更鬆、ranking 訊號可能衰減——這是可實測的 open question。可加一個小型 case study（base vs. instruct checkpoint）[RESULT NEEDED]，跑不了就明確劃入 future work。
- 草稿：

Q1 (full FT / RLHF). The decomposition itself only requires matched architecture/dimensions and is agnostic to how the variant was produced; full FT and RLHF fall under the general form with a nonzero head term (the frozen-head simplification no longer applies). The caveat is tightness: the farther the variant drifts, the less optimal orthogonal alignment becomes, so we expect looser bounds and possibly weaker rank signal. As a first data point, applying PRISM between [base] and [instruct] checkpoints gives [XX] ([RESULT NEEDED]; fallback: "we scope the validated claims to PTQ and frozen-head LoRA and state RLHF/full-FT as an open extension in Sec. [X]").

G3T9-Q2｜Mixed / multi-task fine-tuning？從 original base 到 SFT？

- 草稿：

  Q2 (mixed/multi-task data). Our current evidence covers controlled single-task forgetting; we agree heterogeneous SFT is the natural next test. [If run: one mixed-data LoRA configuration ([datasets]) preserves the diagnostic pattern: [XX] ([RESULT NEEDED]).] [Fallback: we will state this scope explicitly in Sec. [X] and flag multi-task SFT as future work rather than claim coverage we have not tested.]

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4. Reviewer eQL6（3 分，conf 3；Quality=2, Clarity=2）逐點回應

此位是「澄清型」reviewer：兩個低分項（Quality、Clarity）都源自 W2/W3 的理解摩擦。W3 是唯一接近技術性反駁的點，答好有機會同時把 Quality 與 Clarity 拉回 3，是 CP 值最高的一份回覆。

eQL6-W1｜符號未定義就使用（Intro 的 ρ_T ρ_P (1−Ω)）；Section 5 關鍵結果放附錄

- 策略：全盤接受，列具體修訂。
- 草稿：

  W1 (notation; results in appendix). Agreed on both counts. We will (i) define ρ_T, ρ_P (feature energy: the Frobenius/RMS scale of the reference feature matrices) and Ω (the normalized Procrustes alignment, Ω∈[0,1], with 1−Ω the dimensionless shape residual) at first use in the Introduction, with a one-line intuition for each axis; (ii) move [Tables/Figures X, Y — the results Sec. 5 argues from] from the appendix into the main text, and (iii) add a notation table. Thank you for flagging these.

eQL6-W2 / Q1｜為何是 orthogonal transformation？linear/platonic representation hypothesis 如何引出 O(d)？

- 本質：CLARITY / 理論動機。回答要給出「為什麼是 O(d) 而不是任意可逆線性映射」的原理性理由。
- 策略：三個理由。(1) **Gauge** 論證：head 是線性的，任意可逆 A 可被 head 吸收（$W \leftarrow WA^{-1}$），若允許對任意 A 對齊，alignment 會把真實的退化「吸收掉」，殘差失去意義；限制在 similarity transform（rotation×scale）保留內積幾何，殘差才測得到真實形變。(2) 可辨識性：Procrustes 在 O(d)×scale 上有 closed form，得到 exact 的 scale/shape 分解（審稿人

pCi8 也肯定這點）。(3) 實證：相關模型的最優 unconstrained 對齊本來就接近 orthogonal（near-isometric backbone；linear/platonic rep. 文獻用 linear probe 對齊良好）。[VERIFY 與論文 Sec. [X] 推導一致]

- 草稿：

W2/Q1 (why orthogonal alignment). Three reasons, which we will spell out in Sec. [X]. (1) *Gauge:* since the head is linear, any invertible linear map A of features can be absorbed into the head; aligning over all invertible A would let the alignment absorb genuine degradation, making the residual meaningless. Restricting to rotations×scale — the isometry group up to scale — preserves the inner-product geometry the linear head reads out, so the residual measures true distortion. (2) *Identifiability:* over O(d)×scale, Procrustes has a closed-form solution yielding the exact scale/shape split (Prop. [X]); no larger group admits this. (3) *Empirical:* the linear/platonic representation results indicate related models align well under linear maps that are in practice near-orthogonal, which is also what we observe (cf. response to R-pCi8 W6: deviation from orthogonality is small except at extreme quantization). The hypotheses thus justify that a *rotation* suffices for alignment, and the linear head justifies why nothing larger should be allowed.

eQL6-W3 / Q1(前半)｜Shape 項被 $\rho\_T\rho P$ 的量級主導而非 $(1-\Omega)$：極端例子 norms→∞、Ω→1 時 shape 項仍→∞；Table 1 顯示 Ω≈1、$\rho\_T\approx\rho\_P$，所以大的 δ 是 ρ 乘積造成的

- 本質：對分解詮釋的技術性質疑，全場最需要精準回答的一點。
- 分析：Reviewer 的極限例子在數學上沒錯，但前提（feature norm → ∞）在實際比較中不發生：比較的是同一個 **base T** 的多個 **variants P**，ρ_T 固定、ρ_P≈ρ_T 且跨 variant 幾乎不變 ⇒ 跨 **variant** 的排序變異幾乎全部來自 **(1−Ω)**。ρ_TρP 不是 confound 而是正確的量綱/能量尺度：logits 隨 feature norm × head norm 縮放，risk-gap bound 必須攜帶能量單位，(1−Ω) 才是無量綱幾何部分——ρ_TρP(1−Ω) 本質上就是對齊後殘差能量 ‖Z_T − s·Z_P·O‖² 的量級。可從論文既有 Table 1/18 直接算兩個數字釘死：(a) 跨 variant 的 ρ_TρP 變異係數（預期只有幾個百分點）[RESULT NEEDED，從現有表格可算]；(b) shape 項與 (1−Ω) 跨 variant 的 Spearman（預期 ≈1）[RESULT NEEDED]。並提供無量綱化版本 δ/(ρ_TρP) 作為附加報告，證明結論不變。
- 草稿：

W3/Q1 (does the ρ_TρP prefactor, not 1−Ω, drive the shape term?). The limit the reviewer constructs is mathematically valid, but we would like to clarify why it does not arise in — and does not confound — our comparisons. The shape term ρ_Tρ_P(1−Ω) is (up to constants) the *aligned residual energy* ‖Z_T − s·Z_P·O*‖²_F: ρ_Tρ_P carries the physical units (logits scale with feature norm × head norm, so any risk-gap bound must carry this energy scale), while 1−Ω is the dimensionless geometric part. Crucially, every comparison in the paper holds T fixed and compares variants P of the same backbone, so ρ_T is constant and ρ_P is nearly so: across the variants of Table 1, $\rho\_T\rho\_P$ varies by only [XX]% (computable from Tables 1/18), while $1-\Omega$ varies by [XX]× ([RESULT NEEDED]). Consequently the Spearman correlation between the shape term and $1-\Omega$ across variants is [XX] ([RESULT NEEDED]) — the *ranking and diagnostic signal comes from $1-\Omega$*; the prefactor sets units, not the ordering. The reviewer's divergent-norm scenario would require comparing models whose feature energies blow up relative to each other, which does not occur among post-training variants of a shared base (and would itself be a scale-axis pathology PRISM is designed to flag). To make this transparent, we will additionally report the dimensionless normalized score $\delta/(\rho\_T\rho\_P)$ alongside δ in Table [X] — all rankings are unchanged ([RESULT NEEDED]) — and add this discussion to Sec. [X].

eQL6-W4 / Q3｜Regularizer 是否犧牲原下游任務（target task）的表現？

- 本質：S5。stability–plasticity 的標準檢查，必須補表。
- 草稿：

W4/Q3 (target-task performance under regularization). We now report target-task metrics alongside forgetting for all methods (new Table [X]): with the shape regularizer, target-task performance is [XX] vs. [XX] (no reg.) and [XX] (replay), i.e., [state honestly: comparable / a drop of Y% / better] ([RESULT NEEDED], [N] seeds). [If a trade-off exists: we also report the regularization-strength sweep $\lambda\in\{...\}$ tracing the stability–plasticity frontier, showing operating points that retain [XX]% of target gains while reducing forgetting by [XX]%.]

eQL6-Q2｜Sec 5.3「Shape preservation admits two lifecycle-specific instances…」語意不明

- 草稿：

Q2 (Sec. 5.3 sentence). Apologies — the sentence compresses too much. Intended meaning: the *same* shape-preservation objective appears at two points of a model's lifecycle: (i) at PTQ time, Hessian-aware reconstruction (GPTQ-style layerwise objectives) implicitly minimizes shape distortion of the layer outputs; (ii) at fine-tuning time, our differentiable $1-\Omega$ regularizer minimizes it explicitly. We will rewrite the sentence as two explicit clauses and add one line connecting each to its equation.

eQL6-Q4｜Sec 5.5 的 Spearman r_s 是和什麼變數算的？

- 草稿：

Q4 (Sec. 5.5 correlation target). With the empirical risk gap: for each variant we compute [the benchmark-measured degradation $|\Delta R|$ — VERIFY: empirical CE gap or task-metric drop], and r_s is computed between each bound term and this quantity across variants within a (family, benchmark) cell. We will state this explicitly in Sec. 5.5 and in every table caption.

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5. Reviewer 8VrD（4 分，conf 3）逐點回應 — 穩住 advocate

此位最深入且立場最正面，回覆要最完整：他若在 discussion 階段幫你說話，兩個 conf-4 的 3 分才有機會鬆動。

8VrD-W1｜Eq.(1) 定義的是 population risk（期望），證明卻用 finite-sample feature matrices，沒有 generalization 項 → 定理應重述或補 finite-sample 論證

- 本質：正當的理論精確性問題，完全可修，是全場最乾淨的得分點。
- 策略：兩層修訂。(1) 把 Theorem 重述為 **empirical risk gap on the reference sample** 的 bound（診斷用途本來就是作用在固定 reference set 上）。(2) 補 Corollary：在 bounded per-token CE 差（由 bounded feature/head norms 保證，這些量 PRISM 本來就在測）下，以 McDiarmid/Hoeffding 給 population 版本加 $O(\sqrt{\log(1/\delta)/N})$ 濃縮項。[需要實際寫出證明——標準論證，rebuttal 期間可完成；注意 CE 無界性要靠 bounded logits 假設處理，明確陳述該假設]
- 草稿：

W1 (empirical vs. population risk). The reviewer is right, and we will fix the statement precisely as suggested. (1) Theorem [X] will be restated as a bound on the **empirical risk gap over the reference sample** — which is also the quantity our diagnostic use case targets (all experiments evaluate on fixed reference sets). (2) We add Corollary [X]: since per-token CE differences are bounded whenever feature and head norms are bounded (quantities PRISM already measures; assumption stated explicitly), McDiarmid's inequality yields |population gap − empirical gap| ≤ [$C\cdot\sqrt{\log(1/\delta)/2N}$] with probability $1-\delta$, giving the population version at the cost of an additive concentration term. The full statement and proof are in the revised Appendix [X] [we can post the restated theorem in this thread if helpful]. No experimental conclusion changes, since all reported quantities were already empirical.

8VrD-W2 / Q1｜Bound 很鬆（B=266.09 vs |ΔR|=0.3658；B=23.24 vs 0.0002）；要求量化 additive/multiplicative relaxation、說明有無 ordering 以外的 operational threshold

- 本質：S1。他給的兩組數字顯示 slack 非常數（727× vs ~10^5×），回覆必須直面這點。
- 草稿：

  W2/Q1 (looseness; error analysis of bound magnitudes). We now quantify the relaxation directly: across all (family, benchmark, variant) cells we report the slack B/|ΔR| distribution — median [XX], IQR [XX–XX] ([RESULT NEEDED]) — and decompose which inequality in the derivation contributes most at each bit-width ([RESULT NEEDED]; we expect the [simplex-polarization / alignment] step to dominate at [low bits]). The slack is not constant (the reviewer's two cells differ by ~10^2×), which is exactly why we claim only ordering from the raw bound. For an *operational* threshold, we add a held-out isotonic calibration of B → predicted |ΔR| per family: with [K] calibration variants, "B < τ ⇒ |ΔR| < ε" holds with precision [XX] on held-out variants ([RESULT NEEDED]). We will add both analyses (new Sec./Table [X]) and state plainly that uncalibrated B supports ranking only.

8VrD-W3｜Actionability 證據混合：無「只動一個軸」的控制性介入；regularizer 只在 Llama-TruthfulQA 明顯贏；Table 22 有負面 cell（Llama-BBQ +8.6%、Qwen-TruthfulQA +2.7%）；replay 不夠強，要求 SLoRA / two-phase / CLAIM / ArMA；要多 seed、source-task、32-seq 敏感度

- 本質：本篇 review 的核心，也是升 5 分的鑰匙。三個子回應：
  1. 控制性介入（**E4**）：最便宜且最決定性——對同一模型分別注入單軸擾動：(a) 純 scale（features×α 或改 RMSNorm gain）、(b) 純 shape（正交群外的受控線性形變/低秩擾動，保持 norm）、(c) 純 head（只量化/擾動 lm_head）。展示各軸的項選擇性響應（axis i 的擾動只推高 term i）。這直接回應「axes are identifiable」。
  2. **Table 22** 重新詮釋（**E10**，高潛力）：檢查失敗 cell 是否恰好是 PRISM 診斷為「非 shape 主導」的情境。若 Llama-BBQ / Qwen-TruthfulQA 的 forgetting 由 scale/head 軸主導，則 shape regularizer 無效正是 **axis-specificity** 的預測——把「負面結果」翻轉成框架的支持證據。[VERIFY：用論文既有診斷數據核對；若不成立就不要用這個論證，老實承認 mixed]
  3. **Baselines**：加 EWC / L2-SP / feature-KD（canonical，可在 rebuttal 內完成）；他列的四篇（SLoRA、two-phase、CLAIM、ArMA）皆為 2026 年 sequential/continual instruction tuning 方法，若 setting 不同（多任務序列 vs. 單任務 forgetting）要指出差異、承諾在 revision 比對 setting 相符者，至少引用討論。多 seed（≥3）與 source-task 數字併入 E2；32-seq 敏感度併入 E3。
- 草稿：

W3 (actionability; controlled interventions; regularizer baselines; negative cells). (1) *Controlled interventions:* we added the requested single-axis experiments — injecting (a) pure scale change (global feature rescaling), (b) pure geometric distortion (norm-preserving linear shear), (c) head-only perturbation (quantizing lm_head only) into the same base model. Each intervention moves its own term by [XX] while leaving the other two within [XX]% ([RESULT NEEDED]; new Fig./Table [X]). This directly tests axis identifiability under intervention rather than via selected cases. (2) *Negative cells:* we agree Table 22 must be foregrounded, and analyzing it strengthened the paper: in [Llama-BBQ / Qwen-TruthfulQA], PRISM's own diagnosis attributes forgetting primarily to the [scale/head] axis, not shape ([VERIFY from existing diagnostics]) — precisely the regime where a *shape* regularizer should not help. We will present the regularizer as axis-targeted (apply it when the shape term dominates), report all cells including failures in the main text, and state this scoping rule explicitly. [Fallback if diagnosis does not align: report all cells in the main text, characterize when the regularizer helps, and narrow the claim accordingly.] (3) *Baselines:* EWC, L2-SP, and feature-distillation are added under the identical protocol with [N] seeds, with source-task performance and reference-set-size sensitivity reported ([RESULT NEEDED]; new Table [X]). SLoRA, two-phase CIT, CLAIM, and ArMA target sequential continual-instruction settings [VERIFY]; we will add comparisons where the setting matches and discuss all four in Related Work in the revision.

8VrD-W4 / Q2｜r_s=0.831 只基於 Llama cells，Qwen-BBQ 有 −0.34 / −0.66 要在主文誠實呈現；autoregressive corollary 只在 teacher-forced 前綴下成立，需 free-run 實驗或 trajectory-shift 項

- 草稿：

W4/Q2 (negative Qwen-BBQ correlations; free-running). (1) Agreed. The revised main text will report the full correlation matrix including Qwen-BBQ (−0.34, −0.66) with aggregate statistics computed over *all* cells (mean [XX], median [XX] — [RESULT NEEDED]), plus an analysis of the failure: [our current hypothesis + supporting evidence, e.g., BBQ's accuracy metric is weakly coupled to CE for Qwen: [RESULT NEEDED/VERIFY]]. We will not present 0.831 without this context. (2) *Free-running:* on [subset], we recompute PRISM features on model-generated continuations; rank correlation is [XX] vs. [XX] teacher-forced ([RESULT NEEDED]). [Fallback: we will restate Corollary [X] as valid for teacher-forced prefixes only, and add a trajectory-distribution-shift term as an explicitly open extension.] The Limitations section will list reference-set sensitivity, free-run generation, head-varying variants, and feature-extraction cost, as the reviewer suggests.

8VrD-Q3｜Shape regularizer 對 reference set 大小與 domain 的敏感度

- 併入 E3 回覆（G3T9-W2 同款表格 + regularizer 版本）：

  Q3 (regularizer sensitivity). Sweeping the regularizer's reference set over {8, 16, 32, 64, 128} sequences and two domains ([target-domain] vs. [generic C4]) changes retention by at most [XX]% ([RESULT NEEDED]; new Appendix [X]). [Report honestly if sensitive.]

8VrD-Q4｜與更強 forgetting baselines 比較

- 已由 W3-(3) 覆蓋，指向同一張新表。

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6. Rebuttal 期間實驗優先序

優先	**ID**	實驗	回應對象	成本估計	失敗備案
P0	E1	CKA / SVCCA / Procrustes-distance 同場 ranking 比較（用既有 cached activations）	G3T9-W3	低（無需重跑模型）	無備案，必做——核心定位 claim
P0	E2	Regularizer baselines：EWC、L2-SP、feature-KD、**layer-freezing**（AC-C 明列）；≥3 seeds；同報 target-task	AC-C, pCi8-W5, G3T9-W3, 8VrD-W3/Q4, eQL6-W4	中（LoRA 小規模訓練 × 方法數 × seeds）	至少完成 EWC + layer-freezing；其餘 revision 承諾
P0	E3	Reference set ablation：size {8–128}	G3T9-W2, 8VrD-Q3	低	無備案，必做

優先	**ID**	實驗	回應對象	成本估計	失敗備案
		× domain（benchmark vs. C4/WikiText），含 regularizer 版本			
P0	E4	單軸控制性介入（scale-only / shape-only / head-only）	8VrD-W3	低	無備案，必做——decisive
P0	E5	定理重述（empirical risk）+ concentration corollary（純理論）	8VrD-W1	零計算	無
P0	E6	Slack 量化（B/‖ΔR‖ 分佈、逐步驟歸因）+ isotonic calibration	pCi8-W2, G3T9-W4, 8VrD-W2/Q1	低（既有數字分析）	只報 slack 分佈、calibration 留 revision
P0	E10	Table 22 失敗 cell 與 PRISM 軸診斷的對齊分析	8VrD-W3	零計算（既有數據）	若不對齊：誠實 narrow claim
P0-or-narrow	E7	Free-running generation 子集實驗（AC-D 給二選一：實驗或明確劃界；至少跑 1 個 cell）	AC-D, G3T9-W2, 8VrD-W4/Q2	中（需 decoding）	Claim narrowing（corollary 重述為 teacher-forced only + Limitations 明文）——AC 明文接受此備案

優先	**ID**	實驗	回應對象	成本估計	失敗備案
P1	E8	Isometry violation vs. bit-width 曲線	pCi8-W6	低	只做 validity/tightness 澄清
P1	E9	GSM8K answer-span 加權變體	pCi8-W4	低-中	承認結構限制 + 寫入 Limitations
P2	E11	Base→SFT / mixed-data LoRA case study	G3T9-Q1/Q2	中	Scope 劃界，future work

時間配置建議（假設 ~7–10 天 discussion window）：前 15% 完成 E5/E6/E10（零/低計算，先鎖定）與所有實驗 protocol 凍結；60% 跑 E1–E4；20% 寫作與交叉一致性檢查；最後 5% 長度壓縮與 tone 檢查。

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7. 修訂承諾清單（寫進 rebuttal 的「we will」必須逐條落實）

- [ ] (pCi8-W1) Introduction 重寫 contribution statement：量化歸因儀器，非現象發現
- [ ] (S1) 主文明確標示 raw bound 僅支持 ranking；新增 calibration 小節/表
- [ ] (pCi8-W4) GSM8K/long-CoT 移入 first-class limitation + answer-span 分析
- [ ] (pCi8-W6) Validity vs. tightness 澄清 + isometry violation 測量
- [ ] (G3T9-W1) 新增 cost analysis 表 + axis→remediation 對照表
- [ ] (G3T9-W3) 新增 CKA/SVCCA/Procrustes ranking 比較表（主文）
- [ ] (S2) 新增 EWC/L2-SP/feature-KD baseline 表，含 target-task 與 seeds
- [ ] (S4) 新增 reference set ablation 附錄
- [ ] (eQL6-W1) ρ、Ω 首次出現即定義；notation table；附錄關鍵表移主文
- [x] (eQL6-W2) Sec. [X] 補 orthogonal alignment 的三重動機
- [ ] (eQL6-W3) 補 dimensionless normalized score 與 prefactor 討論
- [ ] (eQL6-Q2) 重寫 Sec. 5.3 該句
- [ ] (eQL6-Q4) Sec. 5.5 與所有 caption 明示相關性計算對象
- [ ] (8VrD-W1) Theorem 重述 + concentration corollary（附錄證明）
- [ ] (8VrD-W3) 單軸介入實驗節；regularizer 改為 axis-targeted 敘事（若 E10 成立）
- [ ] (8VrD-W4) Qwen-BBQ 負相關進主文 + 全 cell 聚合統計 + 失敗分析
- [ ] (8VrD) Limitations 補：reference-set 敏感度、free-run、head-varying variants、特徵抽取成本

- [ ] (8VrD-W3) Related Work 補 SLoRA / two-phase / CLAIM / ArMA 討論

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8. 執行注意事項

1. 不可虛構：所有 [XX] 必須是真跑出的數字。任何一個實驗來不及，就用該點附的 fallback 句（narrow concession），寧可少說。
2. 每個 **thread** 自包含：OpenReview 各 reviewer thread 獨立渲染，不要寫「see our response to Reviewer X」；共用的實驗結果各 thread 內各自濃縮一段。
3. 語氣：pCi8 的「convenient choice」不要接招，直接給數字。eQL6 的 W3 回覆開頭先肯定其極限例子數學上成立，再澄清——這位 conf 3、Quality/Clarity 給 2，回覆的清晰度本身就是在修復 Clarity 分數。
4. **Global response** 以 **AC** 五項（**A–E**）為骨架（見 §0.5）：開頭一句「we address each of the five points in the meta-review with new experiments」，A–E 各一段（2–4 句 + 關鍵數字），結尾一句 claim-narrowing 總結。不要用自己的分類法讓 AC 重新對表。
5. 對 **8VrD** 優先投入：他的 W1（理論重述）與 W3（介入實驗）答得漂亮，是從 4 → 5 以及在 AC 討論中獲得 advocate 的最短路徑。8VrD 的關切與 AC 五項重疊度最高，答好他等於答好 AC。
6. 若 E10（Table 22 = axis-specificity 的預測）核對成立，這是整份 rebuttal 最漂亮的一擊——把最刺眼的負面結果轉為框架的支持證據（同時命中 AC-E）；核對不成立就絕對不要用。
7. **AC** 已預告會看「**narrowing of claims**」：主文標題性的 claim（"risk diagnostic"）若在 rebuttal 中被 AC 認定言過其實，會直接坐實 meta-review 的第一條弱點（"weakens the framing"）。建議 rebuttal 主動把定位語言校準為 "a calibrated, label-free ranking-and-attribution instrument, with an explicitly empirical-risk guarantee"——先講清楚它是什麼，再展示它做得到什麼。
8. 時程：meta-review 07-24 才修訂過，AC 正在關注這個 thread；global response 越早掛出（即使部分實驗標註 "results by "）越能影響 AC 的後續修訂。但切記：掛出的每個數字都必須已定案，不掛半成品。